Kinematic and Load-Distribution Analysis of a 4-DOF Parallel Mechanism Built from Paired Coaxial Spherical Five-bar Sets

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Abstract

This paper proposes a Spatial Five-bar Mechanism (SFM): a 4-DOF parallel mechanism built from two Coaxial and Symmetric 5R Spherical Parallel Mechanism (CoSy 5R-SPM) sets that share a common rotation axis. Together with the mechanism, we introduce a kinematic performance framework tailored to its intended use. We first formalize three design goals (balanced vertical load sharing, a wide range of motion, and low end-effector impedance) and define a set of local performance indices that quantify mechanism-level dexterity, torque amplification, and load distribution: the Local Conditioning Index (LCI), the Torque Amplification Ratio (TAR), and a newly proposed Load Distribution Index (LDI). Their relations to the established Global Conditioning Index [20] and Yoshikawa's manipulability [21] are made explicit. The SFM's architecture is then formalized as the radial vector sum of the two CoSy 5R-SPM outputs. This choice lets us derive the forward and inverse kinematics, and the Jacobian, in closed form as the sum of two independent 2-DOF analyses, a significant simplification for a 4-DOF parallel mechanism. Using the resulting Jacobian, we evaluate LCI, TAR, and LDI across the workspace and observe three characteristics: (i) a nearly unrestricted coaxial rotation together with a workspace that reaches roughly 92% of the enclosing sphere, (ii) broadly uniform load sharing among the four actuators, and (iii) the structural elimination of cascaded reflected rotor inertia at the end-effector. Finally, we compare the SFM against a matched 4-DOF serial manipulator and study two application scenarios (an industrial robotic arm and a bipedal robot leg) to illustrate where the mechanism's properties translate into practical value.

Keywords: Parallel mechanism, Spherical parallel manipulator, Coaxial five-bar linkage, 4-DOF manipulator, Kinematic analysis, Jacobian, Performance index, Load distribution, Bipedal robot, Legged robot design.

1 Introduction

Bipedal humanoid robots are no longer confined to research labs. In the last few years a number of commercial platforms (Agility Digit [1], Boston Dynamics' Electric Atlas [2], Unitree G1 and H1, 1X Neo, Figure 02, and Apptronik Apollo) have begun pilot or production deployment in warehouses, manufacturing lines, and service environments, enabled in part by recent advances in Model Predictive Control for dynamic locomotion [3]. As duty cycles lengthen and individual robots repeat the same motions for hours every day over several years, replacing a single component often delays the work of the entire fleet. Matching the expected life of subcomponents, and extending the operational life of the robot as a whole, has therefore become as central to commercial viability as the performance metrics that dominated the research phase.

Among robotic limbs, a bipedal leg is subjected to conditions noticeably harsher than those of a typical manipulator. A single leg must carry and propel the robot's entire body, which pushes designers toward high output density in a light package [4]. Dynamic walking over unknown terrain makes the exact moment of foot contact difficult to predict, so impacts are unavoidable, and the rapid change in the robot's momentum during touchdown loads the drivetrain with intense impulsive forces. Under repeated loading of this kind, reducer fatigue life degrades sharply, which is why bipedal legs tend to require specialized transmissions that general-purpose manipulators do not.

Within a robotic drivetrain, the reducer is typically the component with the shortest service life, and as a consequence it often sets the ceiling on how long the whole robot can operate. The two reducer families most common in bipedal robots (harmonic gears and cycloid gears) have expected lives that scale inversely with the third and ten-thirds power of the applied torque, respectively, and inversely with operating speed [5], [6], [7]. A modest

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fractional reduction in peak torque therefore buys an outsized extension in life, which makes torque reduction a particularly attractive design lever for maintenance cost. Two established strategies exploit this property.

The first introduces a series elastic or viscoelastic element between the actuator and the load to soften impacts, so that the transmission sees a smoothed rather than an impulsive input [8], [9], [10]. The second distributes the end-effector load across multiple actuators through a parallel mechanism [10], [11], [12]; ATRIAS and GOAT are canonical examples, using a coaxial five-bar linkage and a delta-style mechanism respectively to divide both gravitational and impact loads among several active joints. The two strategies are not mutually exclusive, and a hybrid that places a series element on each branch of a parallel drivetrain is entirely possible.

A subset of parallel mechanisms generates only point-centered rotational motion. Of these, the 5R Spherical Parallel Mechanism (5R-SPM) has been used in minimally invasive surgical tools and camera-orienting systems [13], [14], while the 3-DOF Coaxial Spherical Parallel Mechanism (CSPM) has served as a module for mimicking the shoulder and hip joints of animals [15]. The CSPM variants whose input axes are collinear can rotate through that common axis without kinematic interference, allowing an unusually wide workspace for a parallel mechanism, a property that has drawn recent analytical attention [16], [17], [18], [19].

This paper takes that property as a starting point and pushes it in three directions. First, we cast the 4-DOF motion of two coaxial 5R-SPM sets as the radial vector sum of their outputs, which keeps the forward and inverse kinematics and the Jacobian derivable as the sum of two independent 2-DOF analyses. Second, we introduce a Load Distribution Index (LDI) that quantifies how evenly a parallel mechanism shares an end-effector load among its actuators, and we make its relation to the existing Global Conditioning Index [20] and Yoshikawa's manipulability [21] explicit. Third, we compare the proposed SFM against a matched 4-DOF serial manipulator and discuss two application case studies (an industrial robotic arm and a bipedal robot leg) to locate the regimes in which the mechanism's advantages are actually realized.

Paper outline. § 2 formalizes the design goals and the evaluation framework (LCI, TAR, LDI). § 3 presents the SFM's architecture. § 4 derives the forward and inverse kinematics of both the CoSy 5R-SPM building block and the full SFM. § 5 develops the Jacobian and identifies the singularity loci. § 6 uses these indices to evaluate the mechanism across its workspace. § 7 contains the workspace-volume analysis, the serial-manipulator comparison, and the two application case studies. § 8 concludes and lists future work.

2 Design Requirements and Evaluation Framework

2.1 Target Design Goals

We adopt three design goals that capture the durability and performance demands shared by bipedal legs and general-purpose arms.

G1. Balanced vertical load sharing. The dominant vertical loads (landing impacts, gravitational loads) should be split as evenly as possible among the four actuators so that the peak torque seen by any individual reducer is minimized. This is quantified by the LDI defined in § 6.3.

G2. Wide range of motion. The mechanism should offer a large reachable workspace without giving up the structural advantages of a parallel topology, including in principle unbounded rotation about its coaxial axis. We quantify this by the workspace-volume ratio V_R in § 7.1.

G3. Low end-effector impedance. All active joints (motors and their reducers) should sit at the base so that their reflected rotor inertia does not cascade down the link chain to the end-effector [22]. This property follows directly from the SFM's topology, and we discuss it qualitatively in § 7.4.

2.2 Performance Indices

We introduce three local indices whose purpose is to test, pose by pose, whether the mechanism meets G1–G3.

Local Conditioning Index (LCI). Paralleling the Global Conditioning Index of Gosselin and Angeles [20], which integrates the reciprocal of the Jacobian condition number over the workspace, we use the pose-wise reciprocal itself: $\mathrm{LCI}(q) = 1/\kappa(J(q))$, with $\mathrm{GCI} = \int_W \mathrm{LCI}\, dW \,/\, \int_W dW$. The LCI ranges in [0, 1]; values near 1 correspond to near-isotropic behavior and values near 0 to neighborhoods of singularities. Because the SFM couples translational and rotational motion, we evaluate translational and rotational versions, LCI_T and LCI_R, separately in § 6.1. Regions where the LCI collapses coincide with the regions where Yoshikawa's manipulability ellipsoid [21] degenerates.

Torque Amplification Ratio (TAR). For a unit input F (force or moment) applied at the end-effector along a chosen axis, the joint torque vector required to generate it is $J(q)^T F$, and the TAR is the reciprocal of the worst-case component of this vector: $\mathrm{TAR} = 1/\max_i |(J(q)^T F)_i|$. Mathematically the TAR is equivalent to a transmission ratio: it quantifies how much end-effector input (in the force/moment domain) is amplified relative to the largest joint torque it demands. We use the directional decompositions TAR_x, TAR_y, and TAR_z (with F applied along the x, y, and z axes respectively, i.e., $F = \mathbf{F}_x, \mathbf{F}_y, \mathbf{F}_z$) because they expose the way vertical and lateral loads are amplified differently in the SFM, a distinction that matters for bipedal and arm applications (§ 6.2).

3 Architecture of the Spatial Five-bar Mechanism (SFM)

This section gives the geometric picture of the proposed SFM. The building block (§ 3.1) and the pairing rule (§ 3.2) are stated here; Sections 4 and 5 then develop the kinematics and the Jacobian.

3.1 Building Block: Coaxial and Symmetric 5R-SPM

The proposed SFM is assembled from two Spherical Parallel Mechanism (5R-SPM) sets, each with five revolute joints. Figure 1 shows a single set, a Coaxial and Symmetric 5R-SPM (CoSy 5R-SPM), whose kinematics we analyze in § 4.1. The CoSy qualifier refers to two features that simplify analysis: the two input axes are collinear, and the link lengths are chosen symmetrically so that the output attitude decouples into the sum and difference of the two input angles.

3.2 Coaxial Pairing and Radial Vector Sum

In the proposed SFM, two CoSy 5R-SPM sets share a single common rotation axis and sit on opposite sides of the origin, one pointing forward and one pointing backward along that axis. The end-effector position is defined as the radial sum of the two output vectors.

Two consequences follow from this choice. First, because each 5R-SPM set can still be analyzed independently, the full 4-DOF forward and inverse kinematics of the SFM reduces to the closed-form kinematics of a single set, applied twice and summed, a closure property shown in § 4. Second, since all four input axes share the common rotation axis, the SFM inherits the unbounded-rotation property documented for coaxial-input SPMs [16], [17], [18], [19], now extended to four degrees of freedom.

4 Kinematic Analysis

This section follows the standard framework for serial and parallel mechanism kinematics [23]. § 4.1 develops the kinematics of a single CoSy 5R-SPM set; § 4.2 uses these results to obtain the forward and inverse kinematics of the full SFM.

4.1 Kinematics of CoSy 5R-SPM

This section develops the kinematics of the single CoSy 5R-SPM set introduced in § 3.1, using Figure 1 as the reference schematic.

4.1.1 Inverse Kinematics of CoSy 5R-SPM

In Figure 1, the input joint angles are θ_i, and the output attitude is described by θ and Φ, with $i \in \{1, 2\}$ indexing the two input axes. The parameters α and β are the mechanism's design parameters, and u, v, w denote the unit direction vectors of the input, intermediate, and output axes, respectively.

Applying the Spherical Law of Cosines to each input chain gives the identities below.

$$\cos\beta = \cos\alpha\cos\Phi_1 + \sin\alpha\sin\Phi_1\cos\tilde{\theta}_1 \tag{1}$$

$$\cos\beta = \cos\alpha\cos\Phi_2 + \sin\alpha\sin\Phi_2\cos\tilde{\theta}_2 \tag{2}$$

In Eqs. (1)-(2), Φ_i denotes the per-chain input-to-output angle measured from the i-th input axis. The CoSy geometry of Figure 1 relates this per-chain angle to the output pose parameter Φ by $\Phi_i = \pi/2 - \Phi$, so that $\cos\Phi_i = \sin\Phi$ and $\sin\Phi_i = \cos\Phi$. Substituting this identity into Eqs. (1)-(2) and solving for the intermediate angles yields the closed-form expressions below.

$$\tilde{\theta}_1 = \arccos\left(\frac{(\cos\beta - \cos\alpha\sin\Phi)}{\sin\alpha\cos\Phi}\right) \tag{3}$$

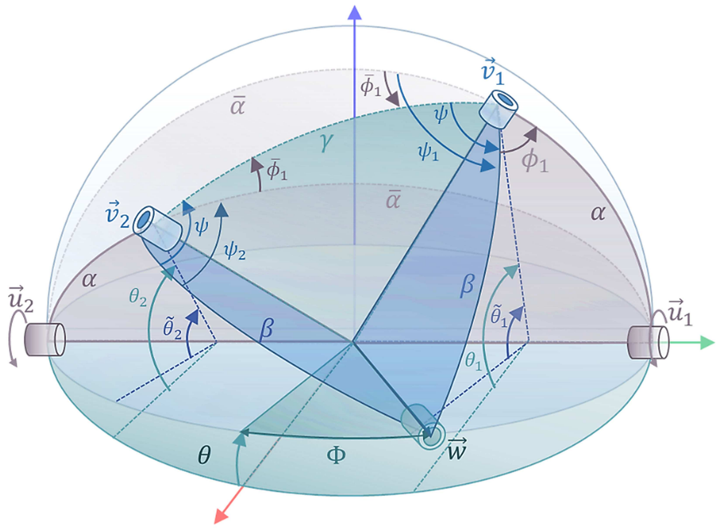

Figure 1: Schematic Diagram of CoSy 5R-SPM

$$\tilde{\theta}_2 = \arccos\left(\frac{(\cos\beta + \cos\alpha\sin\Phi)}{\sin\alpha\cos\Phi}\right) \tag{4}$$

$$\theta_1\left(\theta, \Phi\right) = \tilde{\theta}_1 + \theta = \arccos\left(\frac{\cos\beta - \cos\alpha\sin\Phi}{\sin\alpha\cos\Phi}\right) + \theta \tag{5}$$

$$\theta_2(\theta, \Phi) = \tilde{\theta}_2 + \theta = \arccos\left(\frac{\cos\beta + \cos\alpha\sin\Phi}{\sin\alpha\cos\Phi}\right) + \theta \tag{6}$$

These expressions determine the two input angles θ_1, θ_2 directly from any reachable output pair (θ, Φ).

4.1.2 Forward Kinematics of CoSy 5R-SPM

For the forward map we proceed in the standard sequence: (i) compute the included angle γ between $\mathbf{v}_1$ and $\mathbf{v}_2$ from their dot product, as in Eq. (7); (ii) apply the Spherical Law of Cosines to obtain $\widetilde{\phi}$, as in Eq. (9); (iii) derive ψ from Eq. (10); (iv) use symmetry to write ψ_1 and ψ_2, as in Eqs. (11) and (12); and (v) compose these to obtain the output vector w, as in Eq. (13).

$$\gamma\left(\theta_1, \theta_2\right) = \arccos\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{|\mathbf{v}_1|\,|\mathbf{v}_2|} = \arccos\left(\mathbf{v}_1 \cdot \mathbf{v}_2\right) = \arccos\left(\mathbf{v}_1^T\mathbf{v}_2\right) \tag{7}$$

In Figure 1 the two 5R-SPM chains meet at the output axis via supplementary wedge angles, so for the reverse chain we introduce the supplementary design angle $\bar{\alpha} = \pi - \alpha$, as recorded in Eq. (8). The Spherical Law of Cosines applied to the spherical triangle with sides α, $\bar{\alpha}$, and γ then yields the two equivalent forms on the right-hand side of Eq. (9): the first keeps $\bar{\alpha}$ explicit, and the second follows by substituting $\cos\bar{\alpha} = -\cos\alpha$ and $\sin\bar{\alpha} = \sin\alpha$.

$$\bar{\alpha} = \pi - \alpha \tag{8}$$

$$\tilde{\phi}_1\left(\theta_1, \theta_2\right) = \arccos\left(\frac{\cos\alpha - \cos\bar{\alpha}\cos\gamma}{\sin\bar{\alpha}\sin\gamma}\right) = \arccos\left(\frac{\cos\alpha + \cos\alpha\cos\gamma}{\sin\alpha\sin\gamma}\right) \tag{9}$$

$$\psi\left(\theta_1, \theta_2\right) = \arccos\left(\frac{\cos\beta - \cos\beta\cos\gamma}{\sin\beta\sin\gamma}\right) \tag{10}$$

$$\psi_1\left(\theta_1, \theta_2\right) = \psi + \tilde{\phi}_1 \tag{11}$$

$$\psi_2\left(\theta_1, \theta_2\right) = \psi - \tilde{\phi}_1 \tag{12}$$

$$\mathbf{w} = R_y\left(-\theta_1\right) R_z\left(-\alpha\right) R_y\left(\psi_1\right) R_z\left(-\beta\right) \mathbf{u}_1 = R_y\left(-\theta_2\right) R_z\left(\alpha\right) R_y\left(\psi_2\right) R_z\left(\beta\right) \mathbf{u}_2 \tag{13}$$

Here R denotes a 3×3 rotation matrix and its subscript identifies the axis of rotation.

4.2 Kinematics of the Proposed SFM

This section applies the results of § 4.1 to the full SFM. Let $\mathbf{w}_f$, $\mathbf{w}_b$ be the output vectors of the forward and backward CoSy 5R-SPM sets, with link lengths l_f and l_b. The end-effector position p_e is then the radial vector sum of the two set outputs (Figure 2).

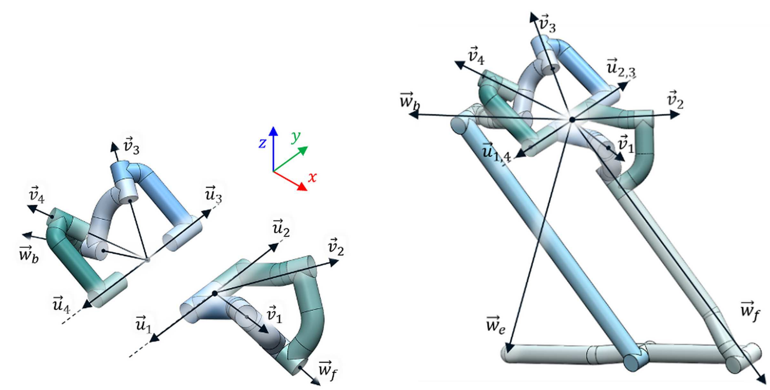

Figure 2: Schematic Diagram (a) Proposed Spatial Five-bar Mechanism (b) Isometric view of SFM (c) Front View of SFM

4.2.1 Forward Kinematics of the Proposed SFM

Using the two front and back link lengths l_f, l_b, the end-effector position is defined as in Eq. (14).

$$\mathbf{w}_e = l_f \mathbf{w}_f + l_b \mathbf{w}_b \tag{14}$$

To express the axial rotation θ_y about $\mathbf{w}_e$, we introduce a local frame on the line from the origin to $\mathbf{w}_e$, with its z-axis along that line and its x-axis through $\mathbf{w}_f$ and $\mathbf{w}_b$. Denoting its axes x_e, y_e, z_e and applying the Z–X–Y Euler rotations relative to the world reference frame gives the relation below. In addition, $\mathbf{w}_m = \mathbf{w}_f$ - $\mathbf{w}_b$ always has the same azimuth as x_e.

Combining these two expressions yields the identity shown in Eq. (18); expanding it gives Eq. (19), from which θ_y is isolated in Eq. (20).

$$\mathbf{x}_e = R_y\left(-\theta_p\right) R_x\left(\theta_r\right) R_z\left(-\theta_y\right) \hat{x} \tag{15}$$

$$\mathbf{w}_m = \mathbf{w}_f - \mathbf{w}_b \tag{16}$$

$$\frac{\mathbf{w}_m}{\|\mathbf{w}_m\|} = \begin{bmatrix} m_x & m_y & m_z \end{bmatrix}^T \tag{17}$$

The remaining Euler angles θ_r and θ_p follow from the standard Z–X–Y decomposition as shown below.

$$R_z\left(-\theta_y\right) \hat{x} = \left(R_y\left(-\theta_p\right) R_x\left(\theta_r\right)\right)^{-1} \frac{\mathbf{w}_m}{\|\mathbf{w}_m\|} \tag{18}$$

$$\begin{bmatrix} \cos\theta_y \\ -\sin\theta_y \\ 0 \end{bmatrix} = \begin{bmatrix} m_x \cos\theta_p + m_z \sin\theta_p \\ -m_x \sin\theta_p \sin\theta_r + m_y \cos\theta_r + m_z \cos\theta_p \sin\theta_r \\ -m_x \cos\theta_r \sin\theta_p - m_y \sin\theta_r + m_z \cos\theta_p \cos\theta_r \end{bmatrix} \tag{19}$$

$$\theta_y = \operatorname{atan2}(m_x \sin\theta_p \sin\theta_r - m_y \cos\theta_r - m_z \cos\theta_p \sin\theta_r, m_x \cos\theta_p + m_z \sin\theta_p) \tag{20}$$

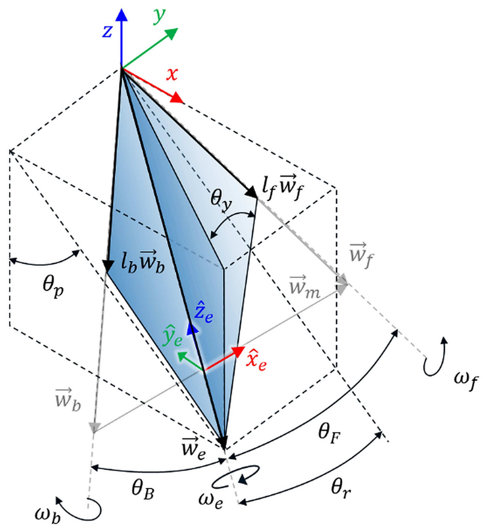

Figure 3: Coordination Definition of Proposed Mechanism

$$\theta_r = \operatorname{atan2}\left(\hat{z}^T \mathbf{w}_e, P_{xz}^T \mathbf{w}_e\right) \tag{21}$$

$$\theta_p = \operatorname{atan2}\left(\hat{x}^T \mathbf{w}_e, \hat{y}^T \mathbf{w}_e\right) \tag{22}$$

4.2.2 Inverse Kinematics of the Proposed SFM

Figure 3 also serves as the diagram for the inverse kinematics. From it, the angles θ_F and θ_B between $\mathbf{w}_e$ and the set outputs $\mathbf{w}_f$, $\mathbf{w}_b$ are obtained as in Eqs. (23)-(24).

$$\theta_F = \arccos \frac{l_f^2 + l_e^2 - l_b^2}{2 l_f l_e} \tag{23}$$

$$\theta_B = \arccos \frac{l_b^2 + l_e^2 - l_f^2}{2 l_b l_e} \tag{24}$$

Given θ_F and θ_B together with the end-effector attitude angles θ_r, θ_p, θ_y derived in § 4.2.1, the individual set outputs $\mathbf{w}_f$ and $\mathbf{w}_b$ are reconstructed through Eqs. (25)-(26).

$$\mathbf{w}_f = -R_y\left(-\theta_p\right) R_x\left(\theta_r\right) R_z\left(-\theta_y\right) R_y\left(-\theta_F\right) \hat{z} \tag{25}$$

$$\mathbf{w}_b = -R_y\left(-\theta_p\right) R_x\left(\theta_r\right) R_z\left(-\theta_y\right) R_y\left(\theta_B\right) \hat{z} \tag{26}$$

The set-level attitude angles Φ_f, Φ_b, θ_f, θ_b are then obtained by the atan2 formulas in Eqs. (27)-(30).

$$\Phi_f = \operatorname{atan2}\left(\mathbf{w}_f^T \hat{z}, \mathbf{w}_f^T P_{xz}\right) \tag{27}$$

$$\Phi_b = \operatorname{atan2}\left(-\mathbf{w}_b^T \hat{y}, \mathbf{w}_b^T P_{xz}\right) \tag{28}$$

$$\theta_f = \operatorname{atan2}\left(\mathbf{w}_f^T \hat{z}, \mathbf{w}_f^T \hat{x}\right) \tag{29}$$

$$\theta_b = -\operatorname{atan2}\left(\mathbf{w}_b^T \hat{z}, -\mathbf{w}_b^T \hat{x}\right) \tag{30}$$

Applying the inverse kinematics of § 4.1 to each set then recovers the four input angles θ_1, θ_2, θ_3, θ_4 in closed form, as shown in Eqs. (31)-(34). The key observation is that a 4-DOF inverse kinematics has been reduced to two decoupled 2-DOF inverse kinematics, a direct consequence of the topology choice made in § 3.2.

$$\theta_1\left(\theta_f, \Phi_f\right) = \arccos\left(\frac{\cos\beta - \cos\alpha\sin\Phi_f}{\sin\alpha\cos\Phi_f}\right) + \theta_f \tag{31}$$

$$\theta_2(\theta_f, \Phi_f) = \arccos\left(\frac{\cos\beta + \cos\alpha\sin\Phi_f}{\sin\alpha\cos\Phi_f}\right) + \theta_f \tag{32}$$

$$\theta_3\left(\theta_b, \Phi_b\right) = \arccos\left(\frac{\cos\beta - \cos\alpha\sin\Phi_b}{\sin\alpha\cos\Phi_b}\right) + \theta_b \tag{33}$$

$$\theta_4(\theta_b, \Phi_b) = \arccos\left(\frac{\cos\beta + \cos\alpha\sin\Phi_b}{\sin\alpha\cos\Phi_b}\right) + \theta_b \tag{34}$$

5 Jacobian and Singularity Analysis

This section develops the Jacobian of the SFM and identifies the poses at which it loses rank. Following the elimination technique used in earlier parallel-manipulator work [24], [25], we solve the full Jacobian as the combination of two single-set Jacobians plus a coupling term that accounts for the radial sum.

5.1 Jacobian of a Single 5R-SPM Set

For a single 5R-SPM set, let $\boldsymbol{\omega}$ denote the angular velocity of its output axis. The kinematic identity that the angular velocities about the three consecutive axes $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ sum to $\boldsymbol{\omega}$ gives, after taking a reciprocal vector on the left, the per-chain system below.

We use subscripts f and b to distinguish the front-pointing set from the back-pointing one, so that $\boldsymbol{\omega}_f$ and $\boldsymbol{\omega}_b \in \mathbb{R}^3$ are the 3D angular velocity vectors of the two per-set output rigid bodies. Their relations to the input rates appear in Eqs. (35)-(38) below.

The skew-symmetric operator $(\cdot)_\times$ used in the matrix forms throughout this section is defined as in Eq. (42).

Rewriting Eqs. (36) and (37) in compact symbolic form yields the following.

$$(\mathbf{v}_1 \times \mathbf{w}_f) \cdot \left(\mathbf{u}_2\dot{\theta}_1 + \mathbf{v}_1\dot{\psi}_f + \mathbf{w}_f\dot{\varphi}_f\right) = (\mathbf{v}_1 \times \mathbf{w}_f) \cdot \boldsymbol{\omega}_f \tag{35}$$

Where, $\Theta = \begin{bmatrix}\Theta_f & \Theta_b\end{bmatrix}^T$, $\Theta_f = \begin{bmatrix}\theta_1 & \theta_2\end{bmatrix}^T$, $\Theta_b = \begin{bmatrix}\theta_3 & \theta_4\end{bmatrix}^T$

Taking the pseudo-inverse on the right and solving for $\boldsymbol{\omega}_f$, $\boldsymbol{\omega}_b$ gives the expressions below.

$$(\mathbf{v}_2 \times \mathbf{w}_f) \cdot \left(\mathbf{u}_2\dot{\theta}_2 + \mathbf{v}_2\dot{\psi}_f + \mathbf{w}_f\dot{\varphi}_f\right) = (\mathbf{v}_2 \times \mathbf{w}_f) \cdot \boldsymbol{\omega}_f \tag{36}$$

$$(\mathbf{v}_3 \times \mathbf{w}_b) \cdot \left(\mathbf{u}_1\dot{\theta}_3 + \mathbf{v}_3\dot{\psi}_b + \mathbf{w}_b\dot{\varphi}_b\right) = (\mathbf{v}_3 \times \mathbf{w}_b) \cdot \boldsymbol{\omega}_b \tag{37}$$

The per-set Jacobians J_f and J_b follow directly, as shown below.

$$(\mathbf{v}_4 \times \mathbf{w}_b) \cdot \left(\mathbf{u}_1\dot{\theta}_4 + \mathbf{v}_3\dot{\psi}_b + \mathbf{w}_b\dot{\varphi}_b\right) = (\mathbf{v}_4 \times \mathbf{w}_b) \cdot \boldsymbol{\omega}_b \tag{38}$$

$$\begin{bmatrix}(\mathbf{v}_1 \times \mathbf{w}_f)\cdot\mathbf{u}_2 & 0 \\ 0 & (\mathbf{v}_2 \times \mathbf{w}_f)\cdot\mathbf{u}_2\end{bmatrix}\begin{bmatrix}\dot{\theta}_1 \\ \dot{\theta}_2\end{bmatrix} = \begin{bmatrix}(\mathbf{v}_1 \times \mathbf{w}_f)^T \\ (\mathbf{v}_2 \times \mathbf{w}_f)^T\end{bmatrix}\boldsymbol{\omega}_f \tag{39}$$

$$\begin{bmatrix}(\mathbf{v}_3 \times \mathbf{w}_b)\cdot\mathbf{u}_1 & 0 \\ 0 & (\mathbf{v}_4 \times \mathbf{w}_b)\cdot\mathbf{u}_1\end{bmatrix}\begin{bmatrix}\dot{\theta}_3 \\ \dot{\theta}_4\end{bmatrix} = \begin{bmatrix}(\mathbf{v}_3 \times \mathbf{w}_b)^T \\ (\mathbf{v}_4 \times \mathbf{w}_b)^T\end{bmatrix}\boldsymbol{\omega}_b \tag{40}$$

5.2 Jacobian of the Full SFM

Let $v \in R^{3\times 1}$ denote the Cartesian linear velocity of the end-effector. The Jacobian of the full SFM is built from the per-set Jacobians of the two 5R-SPM chains, combined through the radial-sum relation. For a column vector $A = [a_1, a_2, a_3]^T$, we use the auxiliary skew-symmetric notation $A_\times$ throughout this section, defined as below.

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}^T \tag{41}$$

$$A_\times = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \tag{42}$$

The kinematic chain equations of each 5R-SPM set (front and back) take the matrix forms below.

$$A_f \dot{\Theta}_f = B_f \boldsymbol{\omega}_f \tag{43}$$

$$A_b \dot{\Theta}_b = B_b \boldsymbol{\omega}_b \tag{44}$$

Solving for the per-set angular velocities yields the expressions below.

$$\boldsymbol{\omega}_f = B_f^+ A_f \dot{\Theta}_f \tag{45}$$

$$\boldsymbol{\omega}_b = B_b^+ A_b \dot{\Theta}_b \tag{46}$$

The per-set Jacobians J_f and J_b that map $\dot{\Theta}_f$ and $\dot{\Theta}_b$ to ω_f and ω_b follow directly.

$$J_f = B_f^+ A_f \tag{47}$$

$$J_b = B_b^+ A_b \tag{48}$$

The end-effector linear velocity v sums the contributions from the two coaxial sets, giving the expression below.

$$v = -l_f \mathbf{w}_f \times \boldsymbol{\omega}_f - l_b \mathbf{w}_b \times \boldsymbol{\omega}_b \tag{49}$$

In matrix form this becomes the relation below.

$$v = \begin{bmatrix} -l_f \mathbf{w}_{f\times} & -l_b \mathbf{w}_{b\times} \end{bmatrix} \begin{bmatrix} J_f \dot{\Theta}_f \\ J_b \dot{\Theta}_b \end{bmatrix} = J_v \dot{\Theta} \tag{50}$$

The Jacobian J_v that relates the input rates $\dot{\Theta}$ to v is therefore defined as below.

$$J_v = \begin{bmatrix} -l_f \mathbf{w}_{f\times} J_f & -l_b \mathbf{w}_{b\times} J_b \end{bmatrix} \tag{51}$$

To separate the rotational component, we define $\mathbf{u}_e$ as the unit direction of $\mathbf{w}_e$ and let $\boldsymbol{\omega}_e$ and $\dot{r}$ denote the rotational and extensional rates along that direction, as below.

$$\mathbf{u}_e = \frac{\mathbf{w}_e}{\|\mathbf{w}_e\|} \tag{52}$$

The extensional rate $\dot{r}$ along $\mathbf{u}_e$ is obtained by projecting v onto $\mathbf{u}_e$, giving the relation below.

$$\dot{r} = \mathbf{u}_e^T v = \mathbf{u}_e^T J_v \dot{\Theta} \tag{53}$$

Using the axial rotation $\boldsymbol{\omega}_e$, the velocities of $\mathbf{w}_f$ and $\mathbf{w}_b$ relative to the end-effector are expressed as below.

$$v_f - v = -\left(\mathbf{w}_f - \mathbf{u}_e\right) \times \boldsymbol{\omega}_e \tag{54}$$

$$v_b - v = -\left(\mathbf{w}_b - \mathbf{u}_e\right) \times \boldsymbol{\omega}_e \tag{55}$$

Subtracting the two and solving for $\boldsymbol{\omega}_e$ yields the following.

$$\boldsymbol{\omega}_e = \left(\mathbf{w}_{f\times} - \mathbf{w}_{b\times}\right)^{-1}\left(v_f - v_b\right) \tag{56}$$

Substituting $J_f\dot{\Theta}_f$ and $J_b\dot{\Theta}_b$ from Eqs. (47)-(48) into Eq. (56) gives $\boldsymbol{\omega}_e$ directly in terms of $\dot{\Theta}$, as shown below.

$$\boldsymbol{\omega}_e = \left(\mathbf{w}_{f\times} - \mathbf{w}_{b\times}\right)^{-1}\left[\mathbf{w}_{f\times}J_f \quad -\mathbf{w}_{b\times}J_b\right]\dot{\Theta} \tag{57}$$

The Jacobian J_e from the input angles to the axial rotation $\boldsymbol{\omega}_e$ is therefore as follows.

$$J_e = \left(\mathbf{w}_{f\times} - \mathbf{w}_{b\times}\right)^{-1}\left[\mathbf{w}_{f\times}J_f \quad -\mathbf{w}_{b\times}J_b\right] \tag{58}$$

Denoting by Ω_e the scalar form of $\boldsymbol{\omega}_e$, the relation reads as shown below.

$$\Omega_e = \mathbf{u}_e^T J_e\dot{\Theta} \tag{59}$$

The total angular velocity $\boldsymbol{\Omega}$ decomposes into the axial component $\mathbf{u}_e\Omega_e$ and a tangential component $\boldsymbol{\Omega}_t$ orthogonal to $\mathbf{u}_e$, as below.

$$\boldsymbol{\Omega} = \boldsymbol{\Omega}_t + \mathbf{u}_e\Omega_e \tag{60}$$

The tangential component $\boldsymbol{\Omega}_t$ relates to v and $\dot{r}$ through the kinematic identity below.

$$v = -\mathbf{w}_e \times \boldsymbol{\Omega}_t + \mathbf{u}_e\dot{r} \tag{61}$$

Solving Eq. (61) for $\boldsymbol{\Omega}_t$ and using the identity $-w_e \times \boldsymbol{\Omega}_t = (-I + u_e u_e^T)v$ gives the relation below.

$$\boldsymbol{\Omega}_t = \mathbf{w}_{e\times}^{-1}\left(-v + \mathbf{u}_e\dot{r}\right) = \mathbf{w}_{e\times}^{-1}\left(-I + \mathbf{u}_e\mathbf{u}_e^T\right)J_v\dot{\Theta} \tag{62}$$

Substituting $\boldsymbol{\Omega}_t$ from Eq. (62) and Ω_e from Eq. (59) into the decomposition in Eq. (60) yields the relation shown below.

$$\boldsymbol{\Omega} = \left(\mathbf{w}_{e\times}^{-1}\left(-I + \mathbf{u}_e\mathbf{u}_e^T\right)J_v + \mathbf{u}_e J_e\right)\dot{\Theta} \tag{63}$$

The Jacobian J_Ω from the input angles to $\boldsymbol{\Omega}$ is therefore given as below.

$$J_\Omega = \left(\mathbf{w}_{e\times}^{-1}\left(-I + \mathbf{u}_e\mathbf{u}_e^T\right)J_v - J_e\right) \tag{64}$$

Finally, the complete input-to-output velocity relation connecting J_v, J_e, J_Ω, and J_r is as shown below.

$$\begin{bmatrix} v \\ \Omega_e \\ \boldsymbol{\Omega} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} J_v \\ J_e \\ J_\Omega \\ J_r \end{bmatrix}\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix} \tag{65}$$

5.3 Singularity Loci

For completeness we identify two singularity families of the Jacobian J derived in § 5.2, following the Type-I and Type-II taxonomy of Merlet [26].

Direct (Type-I) singularity: $\det(J_x) = 0$. These are poses at which an end-effector velocity cannot be produced by any choice of input rates. In the SFM they occur when the two 5R-SPM output vectors are parallel or antiparallel, i.e. near $\theta_f = \theta_b$ or $\theta_f = -\theta_b$. The LCI collapses to zero on these loci, which also form the outer boundary of the contour regions shown in § 6.1.

Inverse (Type-II) singularity: $\det(J_q) = 0$. These are poses at which input rates fail to reach the end-effector. The geometric constraint $\alpha + \beta - \pi/2 = \Phi_C$ of the CoSy 5R-SPM forces these singularities to appear as $\Phi \to \Phi_C$. They correspond to the boundary of each 5R-SPM's workspace and thus to the workspace boundary analyzed in § 7.1.

The two loci are disjoint; inside the workspace the LCI is bounded away from zero in every direction, and the claims G1–G3 of § 2.1 are evaluated entirely on this interior.

6 Performance Evaluation

Using the Jacobian derived in § 5, we evaluate the three indices of § 2.2 across the workspace and test, numerically, whether goals G1–G3 are met. The representative design parameters are $r = 0.5$ m with $l_f = l_b$ (we refer to this case as the symmetric SFM); by the mechanism's symmetry only one octant of the contour plots is shown.

6.1 LCI Analysis

The translational and rotational motions of the SFM separate into two Jacobian blocks, and we evaluate the LCI on each separately, as in Eqs. (66) and (67), where $\|A\| = \sqrt{\mathrm{tr}(A^T A)}$ for $A \in \mathbb{R}^{m\times n}$.

$$LCI_T = \frac{1}{\sqrt{\left\|J_v^{-1}\right\| \left\|J_v\right\|}} \tag{66}$$

$$LCI_R = \frac{1}{\sqrt{\left\|J_\Omega^{-1}\right\| \left\|J_\Omega\right\|}} \tag{67}$$

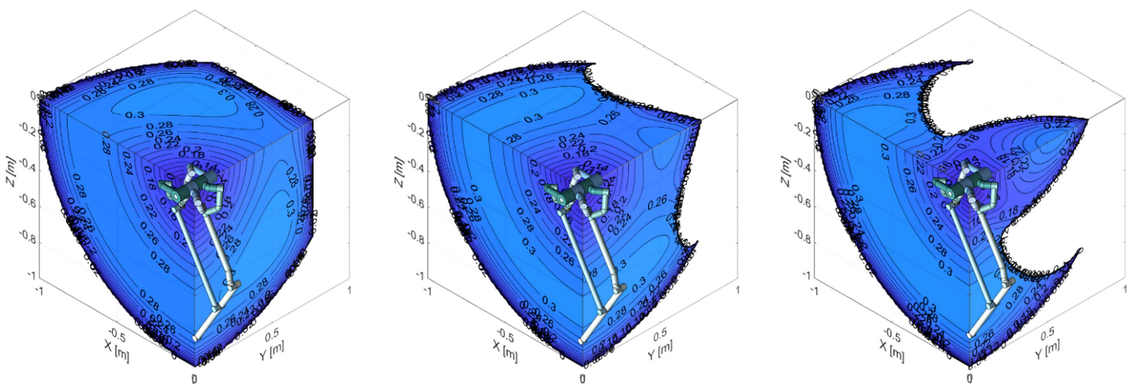

Figure 4: The contour plots shows LCI$_T$ of the SFM when (d) $|\theta_y| = 0°$ (e) $|\theta_y| = 24°$ (f) $|\theta_y| = 45°$

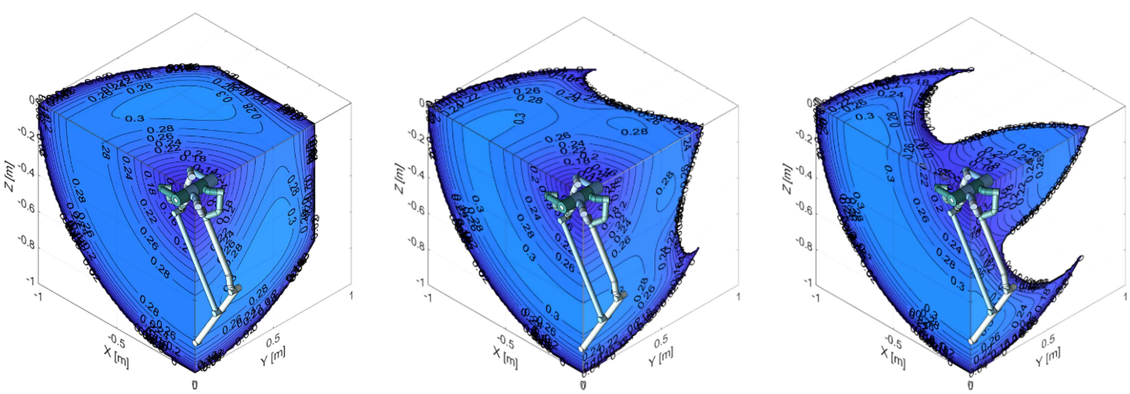

Figure 5: The contour plots shows LCI$_R$ of the SFM when (d) $|\theta_y| = 0°$ (e) $|\theta_y| = 24°$ (f) $|\theta_y| = 45°$

6.2 TAR Analysis

For rotational loads, the four active joints are all revolute, so torque can be amplified through the mechanism. The Torque Amplification Ratio captures this effect: along each axis, TAR$_x$, TAR$_y$, and TAR$_z$ report the ratio of the end-effector rotational output to the worst-case actuator torque, as defined in Eq. (68). Throughout this analysis, joint torques and end-effector inputs are normalized so that the actuator saturation limit equals unity; under this convention, TAR = 1 corresponds to a single actuator carrying its full capacity, while TAR = N (N = 4 for the SFM) corresponds to ideally even sharing among the four actuators.

$$\mathrm{TAR} = \frac{1}{\max_i |(J^T F)_i|}, \quad \|F\| = 1 \tag{68}$$

The TAR shares the symmetry of the SFM across the frontal, sagittal, and horizontal planes, so again only one octant is shown. Because TAR_z can be obtained by rotating TAR_x by 90° about the frontal axis, we omit it. As shown in Figure 6, TAR_y reaches a value of 4 at every point on the sagittal plane and decreases as the pose moves away from that plane. The factor of four comes directly from the SFM's coaxial configuration, in which all four active joints are aligned along the y-axis and their outputs sum. This peak value drops sharply, however, as $|\theta_y|$ grows.

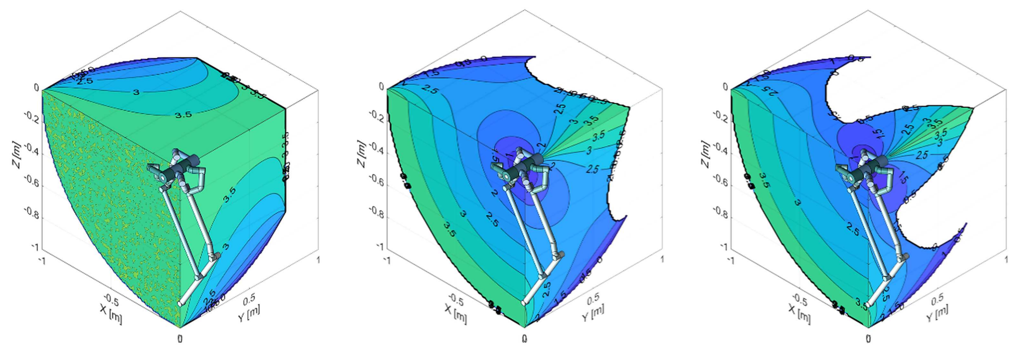

Figure 6: The contour plots shows TAR_y of the SFM when (d) $|\theta_y| = 0°$ (e) $|\theta_y| = 24°$ (f) $|\theta_y| = 45°$

Figure 7 shows TAR_x across the same θ_y range. In the frontal plane TAR_x is small near the origin and grows with distance; in the horizontal plane the behavior is reversed. The same pattern appears in TAR_z with the roles of the planes swapped, and in both cases the distribution becomes noticeably more sensitive to θ_y the further one is from the coaxial configuration.

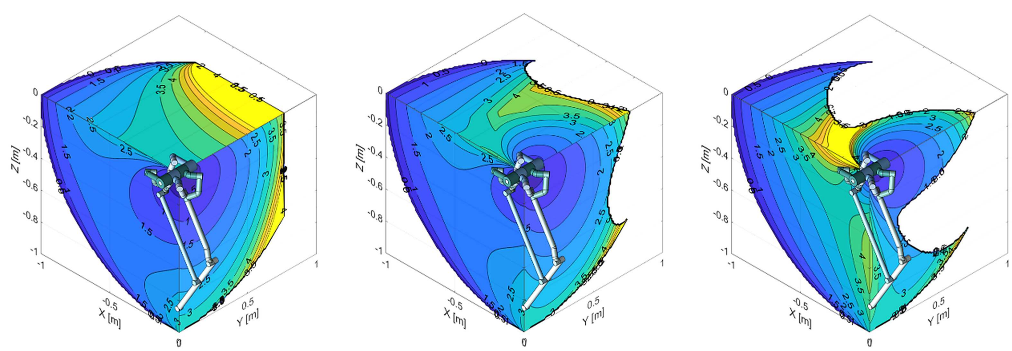

Figure 7: The contour plots shows TAR_x of the SFM when (d) $|\theta_y| = 0°$ (e) $|\theta_y| = 24°$ (f) $|\theta_y| = 45°$

6.3 LDI Analysis

The SFM shares the end-effector load among four active joints; the LDI quantifies the evenness of that sharing. Its directional forms are defined in Eqs. (69)-(70).

Where $\mathbf{F}_y = \begin{bmatrix}0 & 1 & 0\end{bmatrix}^T$, $\mathbf{F}_z = \begin{bmatrix}0 & 0 & 1\end{bmatrix}^T$.

Concretely, for a translational load along a chosen direction, the LDI is the ratio of the minimum to the maximum joint torque magnitude required to hold the pose. By construction, perfect sharing across the four joints maps to 1, while any pose where at least one joint contributes nothing maps to 0; intermediate values quantify how close the bottleneck joint runs to the worst-loaded one.

We refer to the three directional versions as LDI_x, LDI_y, and LDI_z; as with the TAR, LDI_x is omitted by symmetry.

$$\mathrm{LDI}_y = \frac{\min_i |\tau_{y,i}|}{\max_i |\tau_{y,i}|}, \quad \boldsymbol{\tau}_y = J_v^T \mathbf{F}_y \tag{69}$$

$$\mathrm{LDI}_z = \frac{\min_i |\tau_{z,i}|}{\max_i |\tau_{z,i}|}, \quad \boldsymbol{\tau}_z = J_v^T \mathbf{F}_z \tag{70}$$

At $\theta_y = 0$, LDI_z is close to 1 in the neighborhood of the z-axis (Figure 8), meaning that near that line the four joints share the vertical load almost evenly; the sharing quality degrades with lateral distance. Increasing $|\theta_y|$ reduces LDI_z even near the z-axis.

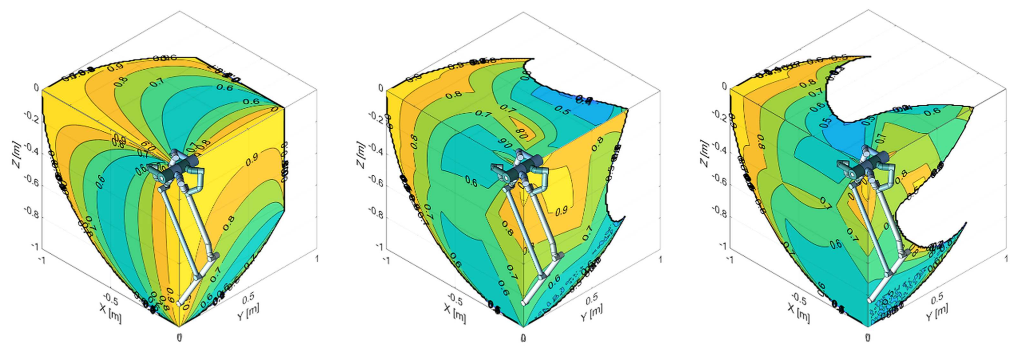

Figure 8: The contour plots shows LDI_z of the SFM when (d) $|\theta_y| = 0°$ (e) $|\theta_y| = 24°$ (f) $|\theta_y| = 45°$

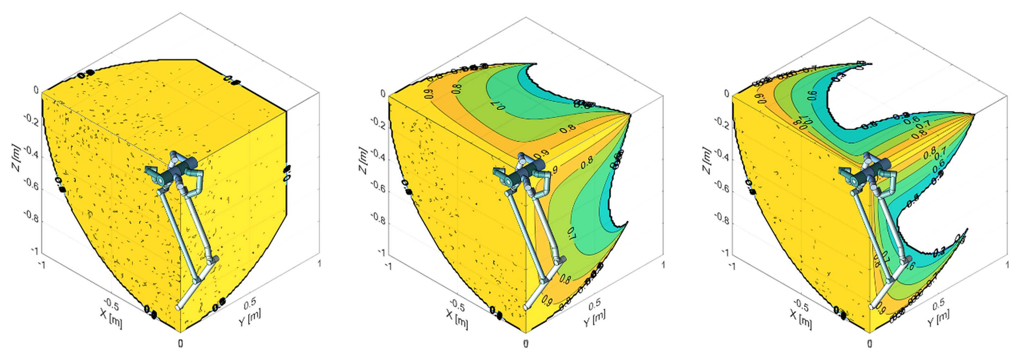

Figure 9: The contour plots shows LDI_y of the SFM when (d) $|\theta_y| = 0°$ (e) $|\theta_y| = 24°$ (f) $|\theta_y| = 45°$

7 Workspace, Comparative Study, and Applications

7.1 Workspace Volume Analysis

Each CoSy 5R-SPM set rotates about the axis perpendicular to the coaxial line by at most α + β - $\pi/2$; we denote this angle Φ_C. For an SFM of radius r, the workspace volume V_{SFM} follows directly from the geometry of Figure 10(a), as in Eq. (71).

The ratio $V_R = V_{\mathrm{SFM}}$ / V_S of the workspace volume to the volume $V_S = \frac{4}{3}\pi r^3$ of the enclosing sphere is then given in Eq. (72).

The Jacobian J_v from input rates to v follows from the block expression below.

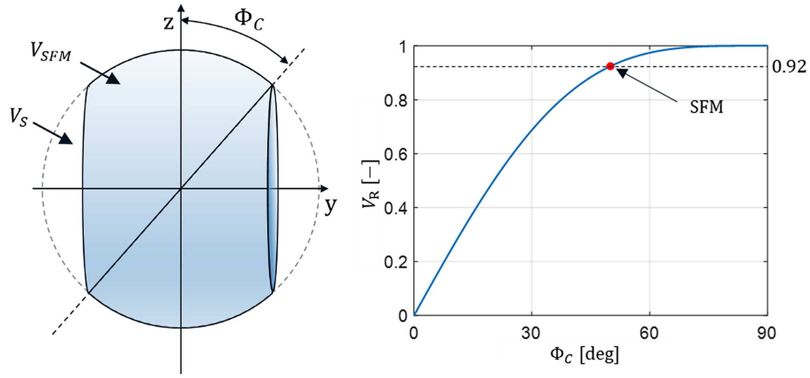

Figure 10: (a) Schematic Diagram describing V_{SFM}, V_S and Φ_C. (b) The V_R of the SFM

$$V_{\mathrm{SFM}} = 2\pi \int_0^{r \sin \Phi_C} z^2 \, dy = 2\pi r^3 \left(\sin \Phi_C - \tfrac{1}{3} \sin^3 \Phi_C\right) \tag{71}$$

$$V_R = \frac{V_{\mathrm{SFM}}}{V_S} = \frac{3}{2} \left(\sin \Phi_C - \tfrac{1}{3} \sin^3 \Phi_C\right) \tag{72}$$

7.2 Comparison with a 4-DOF Serial Manipulator

To quantify the SFM's strengths and weaknesses, we compare it with a 4-DOF serial manipulator (SM) in a Y–X–Z–Y configuration. We chose Y–X–Z–Y because it is the minimal serial configuration providing three-axis positioning together with a yaw degree of freedom; its proximal and distal link lengths are set equal to the SFM workspace radius r = 0.5 m to match the reach. Comparing Figure 11(a) with Figure 5, the SM and the SFM reach similar peak values in translational conditioning, but their high-conditioning regions occupy different parts of the workspace. As the SFM approaches its boundary, LCI_T falls sharply, producing a belt in which the SM exceeds the SFM; inside the SFM's interior, however, the SFM achieves higher dexterity and isotropy than the SM provided the user is willing to restrict some of the workspace and the θ_y range.

In rotational conditioning the picture is reversed: LCI_R of the SM exceeds that of the SFM everywhere (Figure 11(b)). A 4-DOF manipulator, however, is usually augmented with a gimbal wrist when fine rotational control at the end-effector matters, and once the wrist is in place the excess LCI_R of the SM translates into redundant motion that is rarely needed in practice. More importantly, a serial manipulator cannot distribute load across its active joints (it stacks them), so torque amplification in the SFM's sense is unavailable to it.

The net picture is therefore clear: when Cartesian positioning dominates over rotational redundancy and the application demands high payload and a wide range of motion, the SFM offers strengths that the SM cannot match.

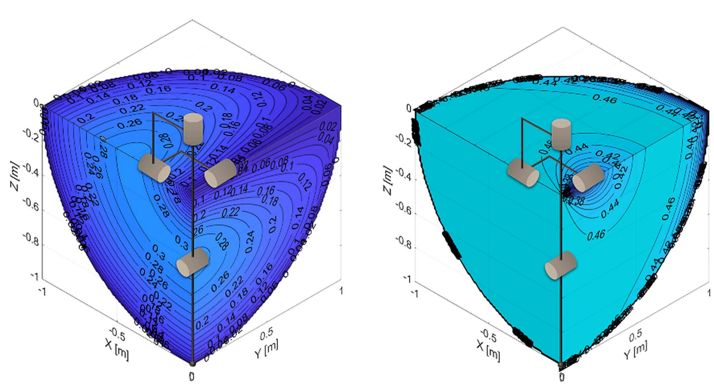

Figure 11: The contour plots describing (a) LCI_T and (b) LCI_R of 4-DOF Serial Manipulator

7.3 Case Study: Industrial Robotic Arm

Several serial industrial arms vertically orient their first joint to work efficiently over a wide horizontal region. The same idea applied to the SFM places the coaxial joint vertically (Figure 12(a)), which gives the gravitational load an even distribution across the four active joints throughout the workspace. The four reducers then age at comparable rates, which enables synchronized replacement cycles and reduces overall maintenance cost. Aligning the coaxial joint with the frontal axis (Figure 12(b)) trades a different set of advantages: a high payload and a tall workspace. In the sagittal plane the four actuators' outputs sum directly along the vertical direction, so each reducer carries only about a quarter of the end-effector load. Under the third-power torque–life relation noted in § 1, that quarter-load translates into roughly a $4^3 = 64$-fold extension of reducer life, an ideal upper bound attained only where the LDI reaches unity (§ 6.3); in practice the effective factor is smaller but still substantial. Both configurations make the SFM a strong candidate for long-duty-cycle robots where maintenance windows are scarce.

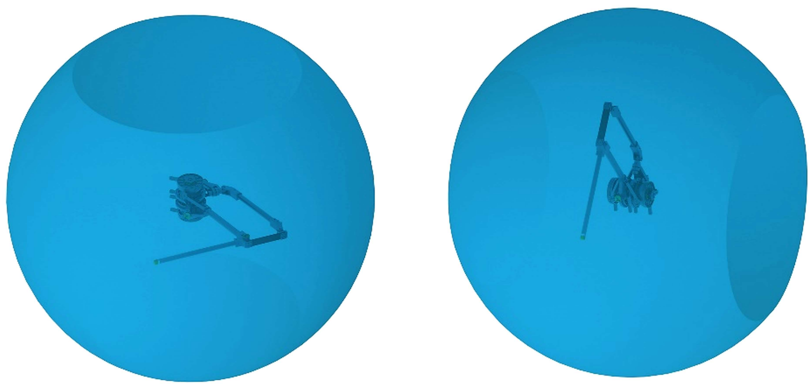

Figure 12: The visualized workspace of robotic arm based on SFM. (a) Coaxial Joint aligned with frontal axis; (b) Coaxial Joint aligned with vertical axis

7.4 Case Study: Bipedal Robot Leg

As control algorithms have matured, bipedal robots now walk dynamically and perform parkour-scale acrobatic motions [27]. These motions ask a leg to produce vertical ground reaction forces and velocities well above body weight, and task-specific motions (climbing over obstacles, for instance) often demand a broad workspace. At the same time, legs do not usually need a wide yaw range, but they do require a dedicated yaw degree of freedom. Each of these demands maps onto a feature of the SFM documented in the previous sections. Beyond the mechanism itself, a pair of SFMs implementing both legs places all eight active joints at the base in a coaxial arrangement. When the robot moves dynamically, this makes the gyroscopic torque of the high-speed rotors both easier to estimate and, within limits, partially self-canceling. Building on these properties, a bipedal lower-body prototype based on two SFMs is currently under construction, and Figure 13 gives a rendering of the intended configuration.

8 Conclusion and Future Work

This paper proposed the Spatial Five-bar Mechanism (SFM), a 4-DOF parallel mechanism formed by two Coaxial and Symmetric 5R-SPM sets that share a common rotation axis, and paired it with a local-index evaluation framework (LCI, TAR, and the newly introduced LDI) whose relation to the established Global Conditioning Index [20] and Yoshikawa's manipulability [21] was made explicit. The radial-sum topology choice allowed the forward and inverse kinematics and the Jacobian to be written in closed form as the combination of two independent 2-DOF analyses, and the same simplicity carried through to the singularity analysis of § 5.3. Evaluating the three indices across the workspace showed that the SFM (i) reaches roughly 92% of the volume of its enclosing sphere in addition to supporting in-principle unlimited rotation about its coaxial axis, (ii) spreads end-effector loads broadly evenly across its four active joints, and (iii) by placing all rotors at the base, structurally avoids the cascaded reflected inertia typical of serial chains. Comparison with a matched 4-DOF serial manipulator and two application case studies (an industrial arm and a bipedal leg) sketched the regimes in which these properties translate into practical advantage. Several strands remain for future work. (a) A dynamic model of the SFM, including the coupling between the two sets, needs to be derived to understand the transient behavior during high-speed tasks. (b) The bipedal prototype currently

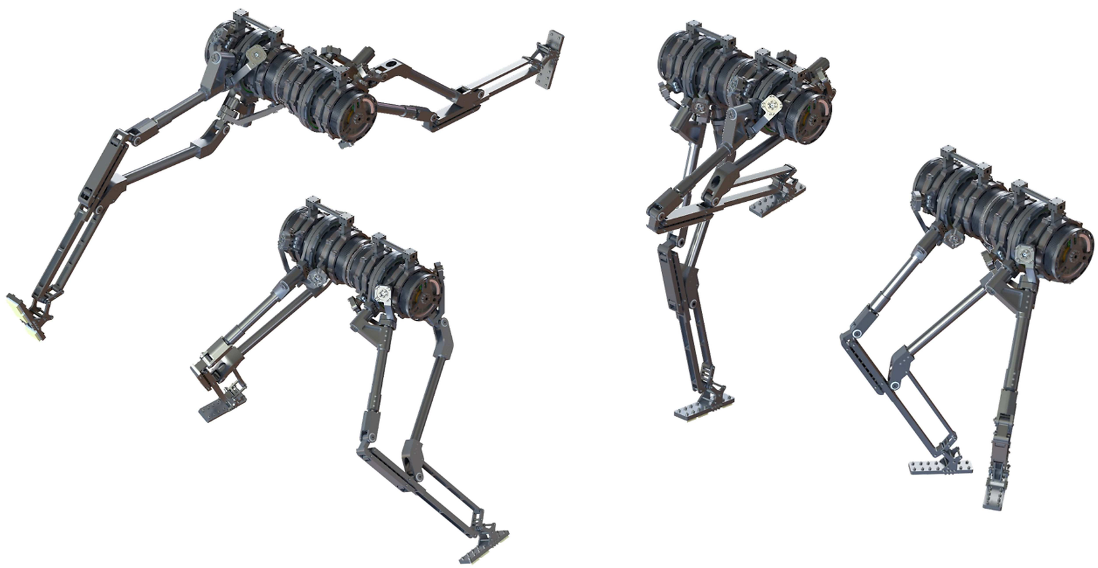

Figure 13: The snapshot of the robotic legs made of proposed mechanism while doing example motion

under construction should enable experimental validation of the claims made here. (c) A sensitivity analysis with respect to manufacturing tolerances and coaxial-axis alignment errors will determine which design parameters deserve the tightest specifications. (d) For the bipedal application, a quantitative analysis of the gyroscopic dynamics induced by the eight coaxially placed rotors would clarify both the control load and the potential for passive cancellation.

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